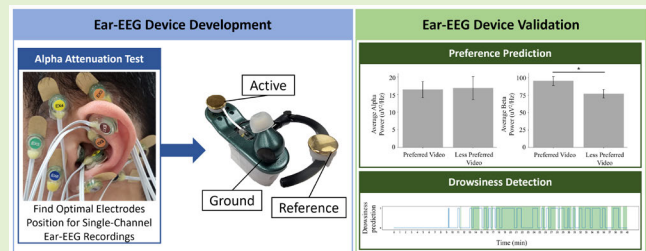


Development of a New Around-the-Ear Electroencephalography Device for Passive Brain–Computer Interface Applications

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Abstract—As interest in passive brain–computer interface (pBCI) technology for everyday applications increases, the development of practical wearable electroencephalography (EEG) recording devices has become increasingly essential. Among the various form factors to implement wearable EEG systems, ear-EEG is frequently employed owing to its usefulness in everyday scenarios. In this study, a new wearable around-the-ear EEG recording device for pBCI applications was developed. The performance of the developed device was validated through two pBCI experiments. During the ear-EEG device design, an alpha attenuation test was conducted to determine the optimal location of a pair of EEG electrodes. The two practical pBCI applications tested in this study were the prediction of users' preferences for short video clips and drowsiness detection during online learning. The experimental results showed an accuracy of 85.71% in terms of preference prediction and a success rate of 80% in terms of drowsiness detection, effectively demonstrating the practicality of the newly developed around-the-ear EEG device for daily life scenarios.

Index Terms—Drowsiness detection, ear electroencephalography (ear-EEG), passive brain–computer interface (pBCI), preference detection, wearable device.



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I. INTRODUCTION

THE brain–computer interface (BCI) enables users to control computers or other external devices using their brain activities [1]. BCI can be broadly categorized into active BCI (aBCI) and passive BCI (pBCI) [2]. aBCI refers to BCI systems that require user's cognitive engagement to control external devices; examples include BCIs based on steady-state visually evoked potentials [3] and motor imagery tasks [4]. In contrast, pBCI typically relies on passive information extraction from brain signals without explicit user input. The examples of pBCI include affective computing systems, which interpret, process, and simulate the users' emotional states, and neuroadaptive systems, which dynamically adjust interfaces reflecting the users' cognitive states. As interest in pBCI increases, many previous studies have presented various pBCI applications, including drowsiness detection [5], neuro-marketing [6], [7], and human–robot interaction [8]. Among the measurement modalities used to implement pBCI systems, electroencephalography (EEG) recorded from electrodes placed on the scalp has been widely used owing to its many advantages, including its noninvasiveness, cost-effectiveness, and high temporal resolution [9], [10].

The conventional EEG recording process in laboratory settings includes the attachment of electrodes to the scalp using a conductive gel to reduce the impedance between the skin and electrodes. Although this approach allows high signal quality, it requires a long preparation time and a cumbersome process to remove the conductive gel from the hair after the recording session, making it difficult to implement practical pBCI applications for everyday use [11]. To address this issue, wearable EEG devices equipped with dry electrodes have been introduced, allowing EEG recordings outside the laboratory [12]. However, conventional wearable EEG devices still require stable attachment of electrodes to the scalp, making it difficult for users to wear the device alone or feel uncomfortable during long-term recording [13].

Ear-EEG is a relatively new form of EEG that employs electrodes located around the ears or inside the ear canal [13], [14], [15], [16], [17]. Compared to traditional headband or headset-type EEG recording devices, ear-EEG systems have several advantages. First, ear protrusions enable users to wear and remove the device readily without assistance from others. Additionally, the absence of hair around the ears or inside the ear canals ensures improved contact between the skin and electrodes, resulting in enhanced signal quality and wearability. Finally, and most importantly, the relatively small size of ear-EEG devices offers a convenient user experience, even during long-term recording, allowing for high utility in various daily life scenarios.

Ear-EEG devices can be categorized into in-ear and around-the-ear EEG devices based on the electrode positions [18]. In-ear EEG devices use electrodes placed within the ear canal or on the ear (e.g., the concha area). Looney et al. [19] pioneered the concept of in-ear EEG by employing personalized earpieces to locate electrodes within the ear canal. Their results demonstrated that in-ear EEG can reliably detect alpha activity and is strongly correlated with conventional scalp EEG. However, personalized earpieces are difficult to employ in practical pBCI applications because of their costly and time-consuming fabrication process. Recently, generic structures and viscoelastic materials have emerged as alternatives to personalized earpieces [17], [20], [21], [22]. For instance, Athavipach et al. [20] proposed a single-channel dry-contact in-ear EEG device designed for the classification of four discrete emotional states, with a silver-adhesive fabric employed as an electrode. This fabric was affixed to the earphone rubber to enhance contact with the surface of the ear canal. Kaveh et al. [17] introduced a wireless in-ear EEG device with a generic design that enabled an improved ear canal contact. Their device consisted of an earpiece with four outward cantilevers that could apply independent pressure to each in-ear electrode and two larger electrodes along the concha cyma and concha cavum as the reference and ground electrodes, respectively.

Around-the-ear EEG devices employ electrodes that are positioned around the ears. This concept was first introduced by Wang et al. [23], who investigated an alternative site for recording high-quality steady-state evoked potential signals from nonhair-bearing regions. Their results showed that the signal-to-noise ratio (SNR) of the behind-the-ear area

was the highest among the many nonhair-bearing regions. Additionally, Debener et al. [24] presented a C-shaped electrode array called cEEGrid, which includes ten electrodes printed on a flexible sheet and arranged to fit around the ear. Based on their work, researchers began to investigate the practicality of around-the-ear EEG devices [25], [26], [27], [28]. Valentin et al. [26] introduced a custom-fit earpiece design with a behind-the-ear piece composed of conductive silicone electrodes. Their device includes a 3-D-printed frame containing a V-shaped folded structure designed to support the behind-the-ear piece against the back of the pinna. This configuration utilizes a spring effect to maintain the pressure on the electrodes, ensuring optimal skin contact. Furthermore, Kaongoen et al. [28] proposed an online BCI system for home appliance control, in which the ear-EEG device was designed in the shape of headphones. This ear-EEG device was composed of flexible silicone electrodes embedded in the cushion of the left and right headphone cups and a biosignal sensing board inside the right headphone cup.

In the previous ear-EEG studies, the number of electrodes varied depending on the design and form factor of the ear-EEG devices [17], [18], [19], [20], [21], [22], [23], [24], [25], [26], [27], [28]. While multichannel ear-EEG devices can enhance signal quality and performance across various applications, they may compromise key advantages of wearable systems, such as usability, comfort, and user convenience. To address this, we developed a dry-contact single-channel ear-EEG device that balances performance and convenience for everyday use. During the development process, an alpha attenuation test was performed to evaluate the quality of ear-EEG signals at various EEG recording locations and identify the optimal site for ear-EEG recording. Based on these findings, we fabricated a single-channel ear-EEG device with an earbud-style design adaptable to any ear shape. To assess the usability of the developed device, we conducted two experiments assuming practical pBCI application scenarios. The first experiment explored the potential of the ear-EEG device to predict user preferences for video clips. The second experiment examined its feasibility for detecting drowsiness during online learning.

II. METHODS

In this study, a new ear-EEG device was designed based on alpha attenuation test results, in which the optimal position of a single-channel EEG electrode for ear-EEG recording was determined. Two participant groups were recruited to validate the usability and feasibility of the ear-EEG device. The first group participated in two experimental sessions to collect ear-EEG signals to develop preference prediction and drowsiness detection algorithms. The other group participated in the same experimental sessions; however, their EEG data were used to validate the developed algorithms.

A. Participants

All participants were university students who volunteered for the experiments. The inclusion criteria were as follows: 1) aged 20 years or above; 2) with no prior history or current

diagnosis of severe depression or other neurological disorders; and 3) had normal or corrected-to-normal vision. Participants who normally wore glasses were asked to wear contact lenses to ensure the proper fitting of the ear-EEG device. The details of the experimental process were explained to each participant before the experiment and all participants provided written informed consent prior to the start of the experiment. After the experiment, the participants received monetary compensation for their participation.

A total of 13 participants (nine males and four females; age: 25.03 ± 2.04 years) volunteered for the preliminary alpha attenuation test. A total of 31 participants (18 males and 13 females; age: 25.03 ± 2.04 years) volunteered for the pBCI application experiment, of which 16 participants (11 males and 5 females; age: 25.43 ± 1.69 years) participated in the preliminary experiments to develop the algorithms; the remaining 15 participants (seven males and eight females; age: 24.60 ± 2.27 years) participated in the main experiments to validate the feasibility of the algorithms.

Among the 16 participants who participated in the preliminary experiments to develop the algorithms for the pBCI applications, one participant's data were excluded from the preference session data owing to severe noise contamination, leaving data from 15 participants (10 males and 5 females; age: 25.25 ± 1.69 years). Additionally, only 12 participants (eight males and four females; age: 25.25 ± 1.69 years) participated in the drowsiness elicitation session because four participants wanted to quit their participation before the drowsiness elicitation session. In the main experiments to validate the algorithms and the proposed ear-EEG device, all participants participated in both experimental sessions; however, data from one participant in the preference session were excluded owing to operator's technical mistakes during the experiment, leaving data from 14 participants for the preference session (six males and eight females; age: 24.79 ± 2.24 years) and data from 15 participants for the drowsiness session (seven males and eight females; age: 24.6 ± 2.27 years).

B. Alpha Attenuation Test

A preliminary alpha attenuation test was conducted to determine the optimal electrode locations for the ear-EEG recordings. The alpha attenuation test is a widely used method to test EEG recording instruments [29] and is based on EEG power suppression in the alpha frequency band (8–12 Hz) when the eyes are open, compared to when the eyes are closed. The alpha attenuation test was employed because the modulation of alpha power between eyes-open and eyes-closed states represents one of the most distinct and robust features detectable in EEG measurements [16], [17], [21]. We assumed that electrodes demonstrating high alpha modulation ratio would also reliably measure EEG signals in other frequency bands, thus ensuring optimal overall signal quality for diverse passive BCI applications. It is worth noting that alpha-band power is also regarded as an effective feature in various pBCI applications, including drowsiness detection [30], [31], [32].

The experiment was conducted with the participants sitting on a chair in front of a liquid-crystal display (LCD) monitor. The distance between the monitor and the participant's eyes

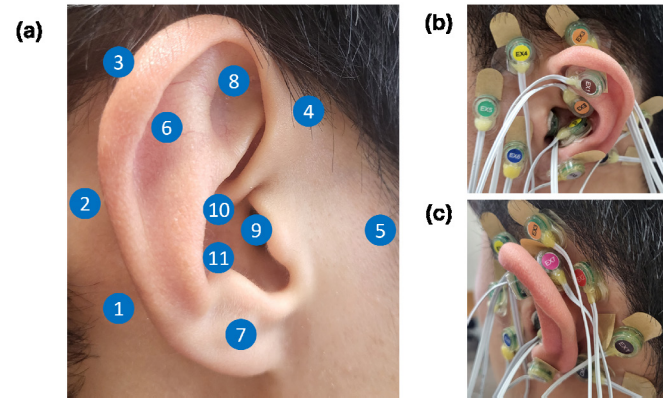


Fig. 1. (a) Illustration of the electrode positions employed in the alpha attenuation test. (b) Side and (c) back views of the attached electrodes.

was set to 60 cm. A black fixation cross was presented in the middle of the display with a white background. The participants were instructed to keep their eyes open for 60 s, followed by a short resting session of 30 s. Subsequently, they were instructed to close their eyes for 60 s.

The ear-EEG data were recorded from 11 electrodes using a commercial biosignal recording system (BioSemi Active Two; BioSemi, Amsterdam, Netherlands) at a sampling rate of 2048 Hz, with the electrode locations, as illustrated in Fig. 1. In Fig. 1(a), the electrode numbers are specified in the blue circles (electrodes 1–5: around-the-ear area; electrodes 6–8, 10, and 11: on-the-ear area; electrode 9: in-ear area). It is worth noting that electrode 6 was attached to the backside of the ear. A common-mode sense (CMS) active electrode and driven right-leg (DRL) passive electrode were used to create a feedback loop for the amplifier reference when the CMS and DRL electrodes were attached to the left and right mastoids, respectively. Then, the recorded ear-EEG signals were downsampled to 128 Hz and bandpass filtered using a fourth-order zero-phase Butterworth filter with cutoff frequencies of 1 and 55 Hz. Additionally, a 60-Hz notch filter was employed to eliminate powerline noise. To determine the best electrode pair (i.e., active and reference electrodes), the recorded ear-EEG signals were rereferenced for all possible electrode combinations, resulting in 55 electrode pairs. The first 10 s of each signal were discarded to prevent noise at the beginning of each recording. Fast Fourier transform was employed to compute the periodograms for 5-s epochs without overlapping. The alpha-band power was computed by averaging the periodogram values in the 8–12-Hz band. The alpha modulation ratio defined in (1) was calculated for all 55 electrode pairs. The alpha modulation ratio is the ratio of the alpha-band powers during the eyes-closed state to the eyes-open state and is defined as

$$\text{Alpha modulation ratio} = a_{\text{eyes-closed}} / a_{\text{eyes-open}} \quad (1)$$

where $\alpha_{\text{eyes-closed}}$ and $\alpha_{\text{eyes-open}}$ represent the alpha-band power during the eyes-closed and eyes-open states, respectively.

To identify the optimal active-reference electrode pair for ear-EEG measurement, Wilcoxon signed-rank tests were performed on all 55 electrode pair combinations to compare

alpha-band power between the eyes-closed and eyes-open states. Among the electrode pairs that showed a significant difference between these states, the optimal pair was selected as a pair with the highest alpha modulation ratio.

C. Ear-EEG Device Development

Our device was designed for comfortable, long-term use in daily life. By employing ear-EEG instead of scalp EEG, the proposed device could enhance wearability, usability, and comfort, with these advantages further maximized by implementing a single-channel system. Optimal electrode positions for the active and reference electrodes were determined based on the results of alpha attenuation tests. For wireless recording of ear-EEG signals, a commercial biosignal recording module (TGAM; NeuroSky, CA, USA) was utilized.

The final design of the ear-EEG device includes one active electrode, one reference electrode, and one ground electrode. The electrode material was determined by comparing the signal amplitudes of Ag/AgCl electrodes and gold plate electrodes. To make the device compatible with various ear shapes, an earbud-style design was employed. Multiple prototypes were fabricated using 3-D printing technology and evaluated for attachment, wearability, and comfort across different users. Based on the evaluations, the final version of the ear-EEG device was developed.

To evaluate the capability to obtain ear-EEG signals, we have assessed the EEG measurement performance of our proposed device relative to a commercial biosignal recording system (BioSemi Active Two; BioSemi, Amsterdam, Netherlands) by comparing alpha modulation ratios derived from data obtained using both devices.

D. pBCI Application Experiments

Experiments on pBCI applications were conducted using the ear-EEG device developed in this study. The experiments consisted of two sessions, referred to as “preference” and “drowsiness” sessions. During the experiments, the participants sat on a chair in a quiet room, where the chair was positioned in front of an LCD monitor. During the experiments, the preference session was conducted first, followed by the drowsiness session. A short break of approximately 5 min was given to the participants between sessions. Each participant was allowed to continue to the next session whenever they wanted within the short break. In each session, video clips eliciting the designated emotions were displayed to the participants. Table I presents the descriptions of the video clips used in the experiments. During the preference session, the participants watched two 300-s video clips; each video clip was designed to induce the specific emotional states of delight and boredom. Video clip 1, for which the target emotion was delight, was created by combining two movie trailers (“Argylle” and “Wonka”). These movie trailers were selected because they had not been released by the time the experiments were conducted. None of the participants had watched the movie trailers before the experiment. In contrast, video clip 2, a recording of a driving test, was relatively static and slow to intentionally induce boredom. Immediately after

TABLE I
VIDEO CLIP DESCRIPTION

Video clip	Length (s)	Description
Video 1 [33, 34]	300	Two short movie trailers from “Argylle” and “Wonka” were combined into a single video clip. The target emotion is delight.
Video 2 [35]	300	Video recording of a driving test. The target emotion is boredom.
Video 3 [36]	2400	Online lecture about perovskite. This video clip was used to induce drowsiness.

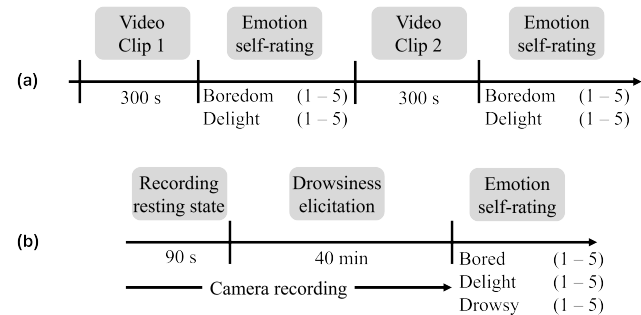


Fig. 2. Timing sequence of (a) preference and (b) drowsiness sessions. Video clips 1 and 2 were shown for 300 s, while the chemistry lecture video clip was presented for 40 min. Emotion self-ratings were conducted using a five-point Likert scale after watching each video clip.

watching each video clip, the participants were asked to assess their current emotional state using a five-point Likert scale (1: low delight/boredom; 5: high delight/boredom). For each participant, the video clip with higher delight and lower boredom was regarded as the preferred video, and the other clip was regarded as the less preferred video. The overall timing sequence of the preference session is shown in Fig. 2(a).

The drowsiness session consisted of three stages, as illustrated in Fig. 2(b). First, the resting-state EEG of the participants was recorded for 90 s. The ear-EEG signals collected at this stage were used as a reference to identify the differences between the drowsy and awake states. After the resting-state stage, video clip 3 was played to the participants for 40 min to elicit drowsiness. An online lecture explaining perovskites was employed. None of the participants majored in chemistry or had never watched a lecture before. The final stage of the drowsiness session was emotional self-rating. At this stage, the participants were asked to assess their current emotional state in terms of two discrete emotions and their level of drowsiness using a five-point Likert scale (1: low drowsiness; 5: high drowsiness). In addition, the participants were asked to describe specific moments when they felt drowsy. During the resting-state and drowsiness-elicitation stages, a video of the front of the upper body of each participant was recorded. Participants’ drowsiness was labeled subjectively by visually inspecting their expressions, such as yawning, blinking, nodding, and eye movement. It is worth noting that all participants consented to recording their faces, and all recordings were deleted after drowsiness labeling.

TABLE II
ACTIVE-REFERENCE ELECTRODE PAIRS WITH
BONFERRONI-CORRECTED $p < 0.05$

Active-Reference Combination	Average Alpha Modulation Ratio	p -value
2 – 5	2.33 ± 1.59	0.04
2 – 4	2.29 ± 1.47	0.04
2 – 10	1.81 ± 0.95	0.03

III. RESULTS

A. Alpha Attenuation Test

The alpha attenuation test was performed with a total of 13 participants using a commercial biosignal recording system. Ear-EEG data were recorded from 11 electrode channels, and 55 electrode pair combinations were assessed through statistical analysis and alpha modulation ratio comparison.

To identify the optimal electrode pair for ear-EEG measurement, statistical analysis was performed using Wilcoxon signed-rank tests to compare the alpha-band power between eyes-closed and eyes-open states. The significance level was set at $p < 0.05$, when a Bonferroni correction was employed for the multiple comparison correction. As the result of the statistical analyses, three electrode pairs showed a significant difference between the eye-closed and eye-open states. The electrode pairs showing Bonferroni-corrected $p < 0.05$ are listed in Table II. Among the three electrode pairs, we selected the electrode pair comprising an active electrode at position 2 and a reference electrode at position 5 as the optimal electrode pair because it achieved the highest alpha modulation ratio (2.33 ± 1.59). Fig. 3 illustrates an example spectrogram of a participant (Participant 2) computed from an ear-EEG signal recorded with the two to five electrode configuration, which clearly shows an increased alpha-band power during the eyes-closed state compared to the eyes-open state. The alpha modulation ratios obtained from three electrode pairs for each participant are provided in Supplementary Table 1.

B. Ear-EEG Device Design

Based on the alpha attenuation test results, electrode positions 2 and 5 were selected as the optimal active-reference electrode combination. Fig. 4(a) shows the developed ear-EEG device. The initial version of the device was fabricated using a 3-D printer, and an ear-round guide made of a flexible material was used to tightly attach the device to the ear. However, as the initial version was relatively heavy and large, it was not appropriate for long-term recording. To address this issue, we developed a second version of the device with a lighter frame and ear hook, as shown in Fig. 4(b)–(d). This device was also 3-D printed, and a plastic ear hook was employed to reduce its weight. A gold-plated electrode was employed because gold electrode showed superior performance over Ag/AgCl electrode in terms of signal amplitude. Moreover, gold electrodes have beneficial characteristics for biosignal measurements, such as being chemically inert and resistant to skin oil, sweat, or both. Additionally, a conductive rubber electrode was used as the ground electrode to maintain tight contact between the electrode and the uneven surface of the concha. The second version of the prototype was used in the main experiments.

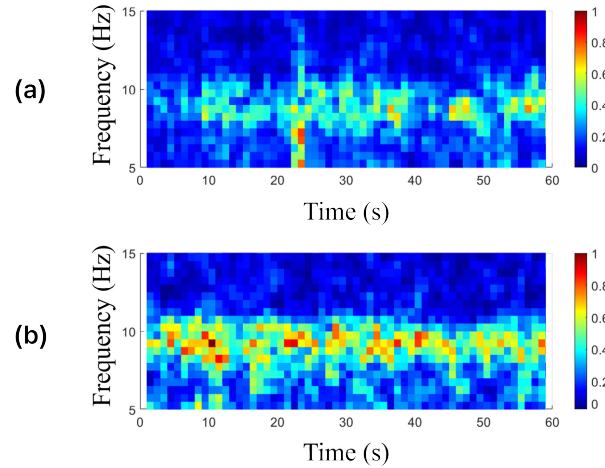


Fig. 3. Examples of EEG spectrograms for Participant 2 in (a) eyes-open and (b) eyes-closed states. The EEG spectrograms are computed from an ear-EEG signal recorded with the two to five electrode configuration. The magnitudes of the spectrograms were normalized by dividing them by the maximum magnitude value observed in the eyes-closed states.

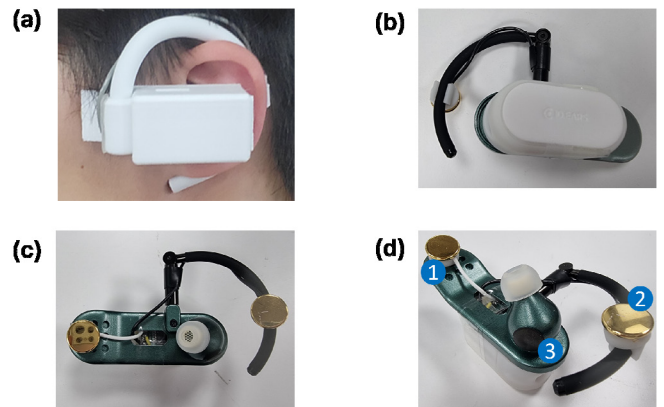


Fig. 4. Fabricated ear-EEG device. (a) Initial prototype. (b) Front, (c) back, and (d) side views of the second prototype. In (d), the electrode numbers are specified in the blue circles: 1–3: active, reference, and ground electrodes, respectively. The second prototype was used in the pBCI application experiments.

TABLE III
COMPARISON OF ALPHA MODULATION RATIOS BETWEEN
COMMERCIAL AND PROPOSED DEVICES

Device	Average Alpha Modulation Ratio
Commercial Device	2.17 ± 1.29
Proposed Device	1.99 ± 0.75

Table III shows the comparison of alpha modulation ratios between the proposed device and a commercial biosignal recording system (BioSemi Active Two). The comparison was conducted for the second participant group, the group whose data were used for validation. The alpha modulation ratio was 1.99 ± 0.75 for the proposed device and 2.17 ± 1.29 for the commercial device. This result demonstrates that EEG signal acquisition performance of the proposed device is comparable to that of the commercial device.

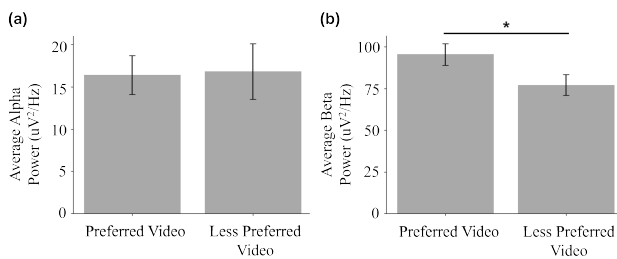


Fig. 5. Average spectral powers of the ear-EEG signals recorded while preferred and less preferred videos were played. The error bars represent standard errors. * indicates $p < 0.05$ in the Wilcoxon signed-rank test. Average (a) alpha and (b) beta power.

C. Preference Prediction Experiment

The preference prediction experiment was conducted using the proposed ear-EEG device. The participants were divided into two groups, where the data from the first group ($N = 15$) were used for algorithm development and the data from the second group ($N = 14$) were used for validation.

In the field of EEG-based neuromarketing, many studies have investigated the neural correlations of human preferences using alpha- and beta-band activities. In particular, Vecchiato et al. [37] found that pleasant and unpleasant advertisements elicited increased theta and alpha activities in distinct brain hemispheres. Similarly, Touchette and Lee [38] demonstrated that frontal alpha asymmetry is closely linked to the unconscious reaction of consumers to product attractiveness. Moreover, Lucchiari and Pravettoni [39] observed that beta-band activity (13–30 Hz) is modulated by a pleasant experience with a favorite brand, and Boksem and Smidts [40] found that high beta activity during movie trailer watching is associated with a high preference for that movie.

Based on previous studies, a preference prediction algorithm was developed by comparing the alpha- and beta-band powers of the ear-EEG signals recorded, while the participants watched two video clips. A 3-s sliding window with a 1.5-s overlap was employed to calculate the absolute alpha- and beta-band powers. To eliminate movement artifacts from the signal, epochs with amplitudes greater than 100 μ V were excluded. Only 90 epochs from the end of each ear-EEG signal were used to average the band powers considering the delay time required to induce the designated emotional states.

Statistical analysis was performed on ear-EEG data from the first participant group ($N = 15$), comparing the average alpha- and beta-band powers between two video clips. Because the normality criterion was not satisfied in the Shapiro–Wilk test, the Wilcoxon signed-rank test was used for statistical analysis. The statistical significance level was set at $p < 0.05$. The average alpha and beta powers for Video clips 1 and 2 are shown in Fig. 5. The Wilcoxon signed-rank test revealed no statistically significant difference ($p \geq 0.05$) between the average alpha powers for a preferred video and a less preferred video, whereas a significant difference ($p < 0.05$) was observed between the average beta powers for two video clips. This indicates that an increase in beta power is closely correlated with the user's preference to the video contents. The effect size, computed using Cohen's d , is 0.4463, indicating a medium

TABLE IV
PREFERENCE PREDICTION RESULTS

Methods	Accuracy (%)
Average Beta Power Comparison	85.71
Bayesian Decision-Making	85.71
Support Vector Machine	92.86
Linear Discriminant Analysis	92.86
Random Forest	92.86

effect. Therefore, user preference was predicted based on the average beta-band power of the recorded ear-EEG signals.

For the second participant group ($N = 14$), the beta-band power was calculated for ear-EEG signals recorded during the two video clips. The preferred video clip was predicted based on the beta-band power averaged over the last 90 moving windows, as in the previous experiment. The average beta powers for each participant are presented in Supplementary Material (Suppl. Table 3).

The preference prediction results of different methods using average beta powers are shown in Table IV. A direct comparison of the average beta powers from the two videos, as well as a Bayesian decision-making approach using the mean and standard deviation of participant group 1, both showed a prediction accuracy of 85.71%. Furthermore, three machine learning classifiers (support vector machine, linear discriminant analysis, and random forest) were trained on the data from participant group 1. These models improved prediction accuracy to 92.86%, demonstrating a notable enhancement over the direct comparison and Bayesian approaches. These results demonstrate that the developed ear-EEG device can be used to predict user preferences for audiovisual content.

D. Drowsiness Detection Experiment

The drowsiness detection experiment was conducted using the proposed ear-EEG device. The participants were divided into two groups, where the data from the first group ($N = 12$) were used for algorithm development and the data from the second group ($N = 15$) were used for validation.

In previous drowsiness detection studies using EEG, an increase in the theta (4–8 Hz) and alpha rhythms was observed in drowsy states [41], [42], [43]. The results from [42] revealed that the alpha and theta power densities increased significantly as the Karolinska sleepiness scale (KSS) score increased. Hwang et al. [43] reported that an overall increase in the theta power in EEG signals was observed during drowsy states. Thus, drowsiness was detected when the sum of the absolute theta and alpha powers exceeded a threshold t_1 . The value of threshold t_1 was determined as double the averaged sum of theta and alpha powers during a 90-s resting state period.

For the first participant group ($N = 12$), it was observed that the frequency of movement noise due to body movement, body and head turning, and eye blinking increased in the drowsy state during the experiments. Therefore, movement noise included in the signal was selected as another feature for drowsiness detection, in addition to the sum of the theta and alpha powers of the ear-EEG signals. The movement

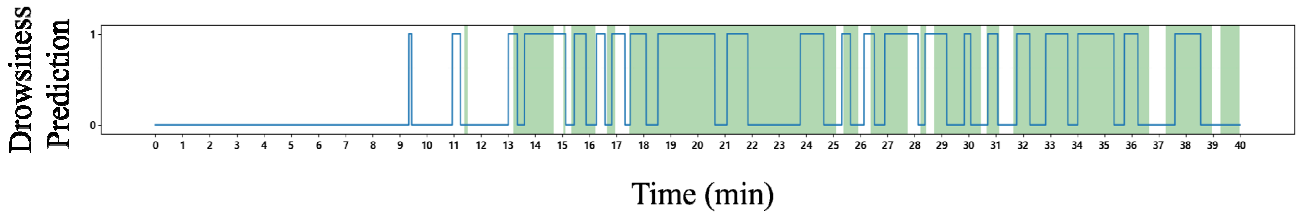


Fig. 6. Drowsiness detection results for Participant 9 in the second participant group of the pBCI experiment. The blue line represents the drowsiness detection result (0: drowsiness not detected; 1: drowsiness detected), and the green shaded bands represent the drowsiness ground truth.

noise frequency was computed by detecting peaks with amplitudes exceeding $100 \mu\text{V}$ using a modified peak detection algorithm [44]. Inspired by the leaky integration and fire models in spiking neural networks, a leaky movement artifact integrator model is proposed to identify the interval in which the movement artifact frequency increases. In this model, variable v starts at zero and continuously decreases by l owing to a persistent leak. In the event of movement artifacts, their frequency was computed and added to v ; in this case, no leaks occurred. The leaky movement artifact integrator model confirmed that the participant was in a drowsy state when v exceeded a threshold t_2 , which was set to 15 in this study. To develop the algorithms, both threshold values were empirically determined based on the ear-EEG signals obtained in the preliminary experiments.

Algorithm 1 Leaky Movement Artifact Integrator Model

```

 $v = 0$ 
for every 0.25 s
   $a \leftarrow$  number of peaks in the 3-s ear-EEG signal
  if  $a > 0$ 
     $v \leftarrow v + a$ 
  else
     $v \leftarrow v - l$ 
  if  $v > t_2$ 
    return drowsy state
  else if  $v < 0$ 
     $v \leftarrow 0$ 

```

Using the EEG power feature and leaky movement artifact integrator model, the drowsiness detection model was configured to identify drowsiness if at least one of the two criteria was satisfied. For real-time operation, a 3-s ear-EEG signal was employed as the drowsiness detection model input for every 0.25 s. The drowsiness detection results using different threshold values for the first participant group are presented in Supplementary Material (Suppl. Table 4). During the optimization process, the threshold was selected at the point that maximized the difference between recall and the false positive rate (FPR), under the condition that recall was greater than FPR.

The main experiments were performed for the second participant group ($N = 15$) using the developed drowsiness detection algorithm. Fig. 6 shows an example of the drowsiness detection results for Participant 9, where the blue line represents the drowsiness detection results (0: drowsiness not detected; 1: drowsiness detected), and the green shaded bands represent the ground truth of drowsiness identified using a camera. Drowsiness detection has several applications. For

TABLE V
DROWSINESS DETECTION RESULTS

Participant No.	Recall	FPR	Success
1	-	0	O
2	0.1255	0.0241	O
3	-	0.2517	X
4	-	0.0183	O
5	-	0.0779	X
6	0.0616	0.0274	O
7	0.1352	0.0052	O
8	0.0088	0.0077	O
9	0.5477	0.1466	O
10	0.3183	0.0568	O
11	0.1714	0.0044	O
12	0.3254	0.0419	O
13	0.4088	0.1848	O
14	0.1419	0.0789	O
15	-	0.1350	X
Mean	0.2245 ± 0.1606	0.0707 ± 0.0736	80 %

example, it can be used to provide alarms when the user is in a drowsy state during driving or studying [45], [46]. Therefore, recall, the proportion of correctly detected actual positive cases, and FPR, the probability of a false alarm, were selected as the performance evaluation indices, as shown in Table V. In this study, recall was computed only if the participants showed drowsiness for at least 1 min. It was judged that the participant's drowsiness was successfully detected when the recall was higher than the FPR. Recall for Participants 1, 3, 4, 5, and 15 was not computed because the drowsy state intervals were less than 1 min. In these cases, drowsiness was successfully detected when FPR was lower than 0.05. For the validation data acquired from 15 participants, drowsiness was successfully detected in all participants except for Participants 3, 5, and 15, resulting in a success rate of 80%. These results demonstrate the feasibility of our ear-EEG device for drowsiness detection. The drowsiness detection results for all participants are presented in Supplementary Material (Suppl. Fig. 1).

IV. DISCUSSION

In this study, a single-channel, dry-contact ear-EEG device was developed with a pair of electrodes positioned around the ear. Only a few conventional ear-EEG studies [16], [17], [47] have explored around-the-ear EEG recording technology, and most studies relied on cEEGrid [22], [24], [27] or manual attachments of conventional EEG recording systems in the around-the-ear area. In this article, a new around-the-ear EEG device with electrode locations determined based on

alpha attenuation test results is proposed. To the best of our knowledge, this is the first study to explore various electrode positions to implement a single-channel ear-EEG system and to validate its feasibility for everyday applications.

The optimal location for the electrode pair for ear-EEG recording was determined to be a combination of electrode positions 2 and 5. Meanwhile, the electrode combinations containing electrode position 9 in the ear canal demonstrated high alpha modulation ratios (Table II). Previous in-ear EEG studies using ear canal electrodes introduced various earpieces, in which the electrodes were in contact with multiple directions of the ear canal [15], [20], [48]. Although these studies demonstrated improved performance using a multichannel in-ear EEG system, our results demonstrate that notable performance in pBCI applications can be achieved with only a single-channel around-the-ear EEG device.

The device developed in this study demonstrated reliable performance in preference classification, suggesting its potential applicability in both research and commercial fields for advancing pBCI techniques. For example, multimodal studies integrating an eye tracker with our EEG device in neuro-marketing research could provide researchers with deeper insights into users' preferences. Additionally, the device could be incorporated into preference-logging applications on edge devices and IoT products. For instance, when a user listens to music on a smartphone or consumes food from a smart refrigerator, the device could classify and predict their preferences for the music or food in real time. This functionality could facilitate timely recommendations for similar content that aligns with the user's preferences.

The proposed ear-EEG device demonstrated the potential for utilization in several pBCI applications; however, it also has limitations that require further improvement. First, individual differences in ear size and ear structure may prevent the electrode from a secure attachment to the skin. Since the proposed ear-EEG device uses dry electrodes, an improper fit could cause signal contamination by artifacts. To enhance more reliable contact performance, future improvements could involve the development of devices in multiple sizes, replacement of the ear hook material with a more flexible one, and utilization of hydrogel-based electrodes. Moreover, since the employment of single-channel EEG in the proposed device could limit its performance compared to multi-channel EEG devices, further investigations are required to explore new form factors that integrate additional electrodes without compromising comfort and usability.

In the pBCI application experiments, the preferred video for each participant was selected based on the emotional self-assessment results after watching the videos. Video clip 1, which was designed to induce an emotional state of delight, was selected as the preferred video clip in most cases; however, Participants 2 and 8 responded that they preferred video clip 2, which was designed to induce boredom (see Supplementary Table 1) in the preliminary experiment. Interestingly, however, the average beta powers of Participants 2 and 8 were higher for their preferred video than for their less preferred video, exhibiting the same tendency as other participants whose preferred video was video clip 1.

The drowsiness detection algorithm was specifically designed for practical applications, particularly in scenarios where an alarm is triggered upon the detection of drowsiness. In such real-world situations, it is essential for alarms to trigger reliably during actual drowsy states rather than producing frequent false alarms during awake states. For instance, in educational settings, it is more beneficial for the system to alert users only when they are falling asleep, rather than disturbing them when they are actively engaged in class. In the drowsiness detection results of the second participant group, there are false negative cases, where intervals labeled as drowsy were not fully detected by the algorithm (see Supplementary Fig. 1). Although optimizing the algorithm to detect all labeled drowsiness intervals could potentially reduce these false negatives, this approach might significantly increase the frequency of alarms, including false alarms during alert periods. Frequent false alarms could lead to inconvenience and decreased responsiveness to alarms, ultimately diminishing the practical effectiveness of the drowsiness detection system. Therefore, the drowsiness detection algorithm was specifically designed to alert when a user is in actual drowsy states, minimizing false alarms. Based on this design consideration, the performance of the drowsiness detection was evaluated using recall and the FPR, resulting in a success rate of 80%. This result demonstrates the feasibility of our proposed device in practical drowsiness detection applications.

The drowsiness ground truth was labeled via visual inspection of the front upper-body view recordings of each participant. Although drowsiness was labeled only when the participant was clearly in a drowsy state, based on the self-rated response after the experiments, it is expected that there were more situations when the participants were in a drowsy state. Therefore, it is possible that the drowsiness detection performance was higher. For drowsiness labeling, previous studies have utilized various labeling techniques, such as signal-based labeling [49], labeling based on sleep stages [50], [51], and the KSS [41], [52]. Based on these previous studies, more accurate labeling using multiple labeling techniques should be introduced in future studies.

In our study, drowsiness labeling was based on subjective observations of participants' external expressions, which could have introduced variability. To preserve the integrity of the experiment, the operator did not inform participants that their drowsiness levels were being monitored. As a result, participants who consciously attempted to stay awake or manage their drowsiness while feeling sleepy may have exhibited neutral facial expressions, leading to their classification as "awake" in such instances. This variability has likely impacted on the accuracy of our drowsiness detection results. To improve the precision of drowsiness detection, future research could incorporate additional objective features from scalp EEG channels, which are recognized as reliable biomarkers for drowsiness states [32]. Moreover, the effectiveness of the system may vary by age, as the current participant sample consisted of individuals in their 20 s. Expanding the age range of participants in future studies would help validate the system's general applicability and effectiveness across a more diverse demographic.

In the pBCI application experiments, beta-band power was employed for preference prediction, while theta- and alpha-band powers were utilized for drowsiness detection. Clarifying the neurophysiological basis for the selection of each frequency band is crucial for a comprehensive understanding of the results. First, beta-band power is generally associated with attention and cognitive processing. Elevated beta activity may reflect increased attention levels when individuals watch preferred content [53]. Moreover, Spitzer and Haegens [54] indicated that beta activity can reflect endogenous information processes involved in the contents that are being viewed. On the other hand, theta-band power typically increases under the conditions of accumulated fatigue or sleep deprivation. Snipes et al. [55] reported increased theta activity associated with sleep deprivation, as well as the occurrence of local sleep, a phenomenon where certain brain regions enter a sleep-like state while others remain awake. Furthermore, alpha-band power increases even during wakefulness when the eyes are closed or external stimuli diminish. Specifically, alpha power may rise as responsiveness to external stimuli decreases during the onset of drowsiness [56].

The preference prediction and drowsiness detection algorithms were developed using the features obtained from a single-channel ear-EEG signal. The proposed methods showed potential for the practical use of wearable ear-EEG devices; however, there is still room for improvement. First, an investigation with more participants in the development of an ear-EEG device may enhance the recording performance of the device. Despite the success of the proposed device in two pBCI applications, additional evaluation of the device with more participants is required to enhance its reliability. Furthermore, with recent advancements in deep-learning technologies, neural networks with relatively low computational costs have been introduced [57], [58], [59], [60]. The performance of the proposed method may be further enhanced using such techniques. An online pBCI application study using efficient deep-learning models on ear-EEG devices will be conducted to further improve the practicality of the developed device. Finally, further exploration is required to expand the application of the proposed device beyond preference prediction and drowsiness detection to include new application areas, such as emotion recognition and attention monitoring. Additionally, integrating the ear-EEG device with other wearable devices, such as smartwatches, could enhance its potential for diverse pBCI applications.

In this study, a new ear-EEG recording system that can be worn for long periods in everyday life was developed. Recently, the worldwide market share of over-the-top (OTT) video services has significantly increased. The prediction of user content preferences using ear-EEG can be used in recommendation systems for OTT video services. Moreover, a drowsiness detection system can assist students by preventing sleepiness while studying and can improve safety while driving by providing an alarm to the driver.

V. CONCLUSION

This article proposed an ear-EEG device designed for pBCI applications, featuring a dry-contact, single-channel

EEG system that prioritizes wearability and comfort for long-term use in daily life scenarios. Electrode positions for the single-channel EEG device were optimized through an alpha attenuation test by evaluating performance variations across different ear-EEG electrode configurations. The performance of the proposed device was assessed in two pBCI applications: preference prediction and drowsiness detection. Algorithms for these applications were developed through preliminary pBCI experiments using data-driven analytics, with the main experiments validating the performance of the algorithms. The results demonstrated a preference prediction accuracy of 85.71% and a drowsiness detection success rate of 80%, highlighting the potential of the device for practical pBCI applications in everyday settings. Future research will focus on expanding its applicability across diverse pBCI environments, paving the way for potential commercialization. Moreover, the earbud-style design, combining discretion and wearability, is expected to offer a competitive advantage in the growing market for wearable EEG devices.

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